Product name: Juramate Cloprostenol Injection 250mcg/ml
Active substance: Cloprostenol Sodium, equivalent to cloprostenol 250mcg
Product description: Clear, colorless solution
Pharmacodynamic: Juramate contains cloprostenol, which is structurally related to prostaglandin F2alpha. Both will cause functional and morphological regression of the corpus luteum (luteolysis) in a variety of species. Luteolysis is then usually followed by a return to oestrus (heat) within 2 to 4 days after treatment. Normal ovulation will usually occur at this oestrus period. Juramate is specific in its luteolytic effect and is therefore ineffective where a functional corpus luteum is not present. A refractory period is usually present for approximately 5 days following normal oestrus when animals will be insensitive to Juramate. Administration at some stages of gestation will result in the loss of the foetus.
Pharmacokinetic: After its administration by injection, cloprostenol is metabolised to acid 9, 11, dihydroxy-15-ceto prost-5-enoic and 9,11, 15-trihydroxyprost-5-enoic which rapidly disappear from the blood, being excreted via the urine in 5-6 hours. Radiolabeled studies show blood levels between 0.0014 and 0.002 :g per ml at 20 minutes – 2 hours after its administration. Subsequently, blood levels fall rapidly, having an apparent half-life of 1-3 hours, falling below 0.00004 :g/ml at 8 hours. No significant concentrations are found at 24 hours in the liver, muscle, heart, kidneys, uterus, ovaries, skin, brain and fat, or in milk after 4 hours.
Indication

Cattle	Horses
Suboestrus (or non-detected oestrus)	Induction of Luteolysis
Termination of Normal but Unwanted Pregnancies	Termination of Pseudopregnancy
Termination of Abnormal Pregnancy -Hydrops of the Foetal Membrane -Removal of mummified foetus	Treatment of Lactation anoestrus
Chronic endometritis (Pyometra)	Establishing oestrus cycle in barren and maiden mares
Induction of Parturition	Synchronisation of oestrus
Ovarian Luteal Cysts	Nomination of the time of service for mares for breeding management
Controlled Breeding Programmes in Cattle	

Dosage & Administration:
CATTLE: Single or repeat doses of 2ml (500µg cloprostenol) by intramuscular injection
1. **Therapeutic**
A single intramuscular dose of JURAMATE is likely to be highly effective in the following clinical conditions of the cow.

Suboestrus (or non-detected oestrus - NDO): This condition occurs in heavy yielding cows, usually at peak lactation, which have normal ovarian cyclicity but in which behavioural manifestations of oestrus are either very mild, transient or absent. Such animals can be treated with JURAMATE following diagnosis of a corpus luteum by rectal palpation and then closely observed for oestrus. Those showing heat should be inseminated. Some animals may have been treated during the refractory period of the cycle and therefore will not respond. Animals not showing heat should receive a further single injection 11 days after the first, and be inseminated 72 and 96 hours later.
Termination of normal but unwanted pregnancies: Pregnancy can be terminated from one week after conception until the 150th day of gestation. Before 100 days gestation, abortion can be induced rapidly and efficiently. Between 100 and 150 days of gestation, results are less reliable, probably because a proportion of cattle may become progressively less dependent upon the corpus luteum for the maintenance of pregnancy. Abortion should not be induced after day 150 of gestation. Treated animals should be kept under supervision until expulsion of the foetus and the placental membranes is complete, as an occasional animal may develop metritis following abortion. Most cows will abort in 3-5 days; if an animal has not aborted by the eighth day, a second injection should be given.
Termination of abnormal pregnancy.
Removal of mummified foetus: Death of the foetus may be followed by its dehydration and degeneration. Induction of luteolysis at any stage of pregnancy will result in the expulsion of this mummified foetus from the uterus into the vagina. Manual removal from the vagina may be necessary. Normal cyclical activity should then follow.
Hydrops of the foetal membranes: Pathological accumulation of placental fluids - hydramnios or hydrallantois - can cause severe physiological complications and death. Surgical drainage is not usually successful in alleviating the condition. A single dose of JURAMATE may be used to induce parturition in such cases. Success has been achieved as early as the sixth month of pregnancy.
Chronic endometritis (pyometra): Damage to the reproductive tract at calving, or post-partum retention of the placenta, frequently lead to infection and inflammation of the uterus. Acute or sub-acute endometritis occurring shortly after parturition may require both local and systemic antibiotic treatment and this often results in resolution of the condition. However, under certain circumstances the endometritis may progress until, at a few weeks post-partum the uterus is very swollen, of a soft doughy consistency and full of purulent discharge. This is characterised by a lack of cyclical oestrus

behaviour and the presence of a persistent corpus luteum. This condition can be successfully treated by induction of luteal regression. Where necessary, treatment may be repeated at 10 to 14 day intervals.

Induction of parturition: Induction of parturition should take place as close to the predicted calving date as possible, and not more than 10 days before. Induction should not be attempted before day 270 of gestation, measured from the confirmed day of conception. All treated animals must receive adequate supervision. In common with all other methods of shortening the gestation period, a higher than usual incidence of retention of the foetal membranes is to be expected. It is now recognised that there may be a reduced survival rate in calves born as a result of early induction. Any increased mortality is due to lack of viability as a result of prematurity, rather than any effect from the prostaglandins.

Ovarian luteal cysts: Where cystic ovaries associated with persistent luteal tissue and absence of oestrus are diagnosed, JURAMATE has proved effective in correcting the condition and brought about a return of cycling. Accurate diagnosis is essential if completely satisfactory results are to be achieved.

2. Controlled breeding programmes in cattle: The luteolytic activity of JURAMATE can be harnessed to control the breeding patterns of cattle. A variety of treatment regimes exists from which it is possible to choose the most appropriate for the characteristics and objectives of each particular individual, group or herd. JURAMATE can be used to complement oestrus detection input or animals may be bred "on schedule" during critical times of the breeding season, without reference to oestrus detection. Use one of the following programmes:

<table>
<tr><td>A</td><td>Day 1: JURAMATE injection</td><td></td><td>Day 11: JURAMATE injection</td><td rowspan="4">BREED:
*At the usual time, relative to detected hear
OR

* At about 72 hours and 96 hours post-injection
OR

Mass AI at 72 hours, with re-insemination of those in oestrus over the next 2 to 3 days</td></tr>
<tr><td>B</td><td>Palpate</td><td>All animals with corpus luteum JURAMATE injection</td><td></td></tr>
<tr><td>C</td><td>Day 1: JURAMATE injection</td><td>Detect heat and breed</td><td>Day 11: All animals not bred JURAMATE injection</td></tr>
<tr><td>D</td><td>Detect heat for 6 days and breed</td><td></td><td>Day 6: All animals not bred JURAMATE injection</td></tr>
</table>

Note: *Best results are obtained where heat detection is utilised. This is generally assisted by the use of tail paint.
*Identification of animals is important.
*Conception rate may be about 20% less if insemination is carried out en masse 72 hours after injection with no following insemination.
*JURAMATE can also be used in systems which include other treatment regimes, e.g. CIDR.

3. Dairy herd: To control oestrus in the individual animal, giving better control of the calving index by allowing artificial insemination without oestrus detection. The number of cows culled as barren is consequently reduced.

To synchronise oestrus in groups of cows to promote management of the herd in groups of suitable size for feeding, A.I., and "drying off". The chances of maintaining a strictly seasonal calving herd are improved, and the number of barren cows at the end of the breeding programme is reduced.

To permit the use of A.I. in dairy heifers which, in turn, allows the speeding up of the breeding programme, the use of a bull known to produce few dystocia problems, better control of heifer management, and "steaming up" prior to calving.

4. Beef herd: To facilitate the use of A.I., improving the progeny through the use of genetically superior bulls. The problems of oestrus detection are avoided and the labour involved in carrying out an A.I. programme is reduced by allowing groups of cattle to be presented instead of single animals.

To permit better management at conception and calving: The calving pattern is altered, resulting in greater average age and weight of calves at weaning. The peak calving period can be forecast more accurately in relation to other events in the farm calendar and there is an improved potential for "flushing" the cows prior to A.I.

HORSES: A single intramuscular injection of 0.5 to 1.0ml for animals up to 400kg body weight, or 1.0ml to 2.0ml for horses weighing 400kg and over.

Induction of luteolysis following early foetal death and resorption. About 8 to 10 percent of all mares which conceive lose the conceptus during the first 100 days of pregnancy. Persistence of luteal function in the ovary precludes an early return of oestrus. Treatment before day 45 is recommended. After that time, no response may be obtained due to the presence of circulating P.M.S.G.

Termination of pseudopregnancy: Some mares covered at normal oestrus and subsequently found to be empty (but not having lost or resorbed a conceptus) display clinical signs of pregnancy. These animals are said to be "pseudopregnant".

Treatment of lactation anoestrus: Failure of lactating mares to cycle again for several months after exhibiting an early "foal heat" can be avoided in most cases. Some variability does however occur.

Barren and maiden mares: Some of these animals will be found, on examination, to have a functional corpus luteum and are either suffering from abnormal persistence of luteal function or are simply failing to exhibit normal oestrus behaviour ("silent heat") while ovarian cyclicity continues.
Synchronisation of oestrus: Under some circumstances this may be a desirable objective in stud management. Treatment during dioestrus usually induces oestrus after 2 to 4 days, with ovulation occurring 8 to 12 days after treatment.
Nomination of the time of service: Mares may be brought into oestrus at will, simply as an aid to successful and economic management of stallions during the breeding season.

Contraindications: None
Interaction with other medicament: None known
Warning & Precautions: Care should be taken when handling this product, especially by women of child bearing age and also by asthmatics. In case of accidental spillage on skin, wash immediately with water.
Pregnancy & Lactation: Do not administer to pregnant animals unless the objective is to terminate pregnancy.
Adverse effect: In cattle, the only apparent effect was mild and transient diarrhoea. In horses, sweating, increased respiratory and cardiac rates, signs of abdominal discomfort, watery diarrhoea, and depression. However adverse reactions are usually mild and transient.
Overdose and Treatment:
Cattle: - At x5 to x10 overdose the most frequent side effect is increased rectal temperature. This is usually transient, however, and not detrimental to the animal. Limited salivation may also be observed in some animals.
Horses: The most frequently observed side effects are sweating and decreased rectal temperatures. These are usually transient, however, and not detrimental to the animal. Other possible reactions are increased heart rate, increased respiratory rate, abdominal discomfort, locomotor incoordination and lying down. If these occur, they are likely to be seen within 15 minutes of injection and disappear within 1 hour. Mares usually continue to eat throughout.

Withdrawal period: Meat: 1 day. Milk: Nil

Dosage form and packaging: 20ml Injectable

Manufacturer:
Jurox Pty Limited
85 Gardiner Street, Rutherford, NSW 2320, Australia

Marketing authorization holder:
QL Pacific Vet Group Sdn. Bhd
No 886-C8, Jalan Subang 9, Taman Perindustrian Subang,47600 Subang Jaya,Selangor, Malaysia

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